

Olasentra Event Operations Command Platform

Phase 16 Implementation Package

Database Optimization — Full deploy blueprint

v1.0 · 6 June 2026 · SQL artifacts generated · PHP not modified

Await Gate F approval before production execution



1. Objective & Scope

Item	Detail
Phase	P16 — Database Optimization
Business value	Faster ops queries; referential integrity; migration runner completeness; schema foundation for P18–P22 platform modules
In scope	Runner consolidation manifest, indexes, staff_id backfill, FK constraints, 8 platform tables, diagnostic SQL
Out of scope (Phase 16b)	PHP runner update, deleteStaffProfileCompletely extension, UI for Cleanup Center
Blocked until complete	Phase 15 Security Hardening (per approval directive)

2. Current Database Analysis

2.1 Existing tables (20 production tables)

Table	Rows (typical)	Phase 16 action
events	32+	Add index idx_events_active_date
staff_registrations	36+ (event 1)	Index + backfill staff_id
staff	~35 unique	Index email; backfill target
attendance	Low	No schema change
staff_messages	Low	Add FK → staff
app_notifications	Growing	Add inbox index
saved_job_records	Low	Add FK event_id, created_by
personal_invoice_lines	Low	Add FK → saved_job_records
commission_invoices	Low	Already has FKs
admin_users, system_settings, venues, work_types, staff_blacklist, push_subscriptions, email_reminder_log, admin_audit_log, website_visits, ip_geo_cache	—	No change

2.2 Migration runner gap (critical finding)

includes/database-migrations.php runs only phases **3–34 + 43**. Missing from runner but SQL files exist:

```
migrate-phase11-site-urls.sql
migrate-phase12-company-home.sql
migrate-phase13-website-content.sql
migrate-phase30-staff-role-varchar.sql
migrate-phase31-staff-registration-save.sql
migrate-phase35-staff-table.sql ← staff table may exist only via manual run
migrate-phase36-populate-staff-table.sql
migrate-phase37-add-staff-id-to-registrations.sql
migrate-phase38-link-registrations-to-staff.sql
migrate-phase39-staff-profile-token.sql
migrate-phase40-psa-licence-fields.sql
migrate-phase41-force-profile-update.sql
migrate-phase42-website-visits.sql
migrate-phase44-staff-messages.sql
migrate-phase45-job-records.sql
migrate-phase46-job-record-type.sql
migrate-phase47-personal-invoice-lines.sql
```

Additionally **24 ensure*Schema()** functions in PHP create tables at page-load — Phase 16b will reduce reliance after runner applies SQL.

2.3 Known integrity gaps (audit)

- staff_registrations.staff_id nullable — some rows unlinked
- staff_messages.staff_id — no FK
- personal_invoice_lines.invoice_id — no FK
- saved_job_records.event_id — no FK
- deleteStaffProfileCompletely() — does not delete staff_messages, app_notifications, push_subscriptions (Phase 16b PHP)
- Queries use LOWER(email) — index on staff.email helps; full normalize deferred

2.4 Production baseline (must not change counts)

```
Event #1 Nick Cave: expect 36 registrations (35 approved, 1 rejected)
api/health.php: app, database, storage = ok
```

3. Exact Files Affected

3.1 SQL artifacts (GENERATED — ready to run)

File	Run order	Mutates data?
database/phase16-precheck.sql	Step 2	Read-only
database/migrate-phase35-..47.sql	Step 4 (if tables missing)	Yes (if not applied)
database/migrate-phase48-composite-indexes.sql	Step 5	No
database/migrate-phase49-backfill-staff-id.sql	Step 6	Yes — links/creates staff
database/migrate-phase50-foreign-keys.sql	Step 8	No (constraints only)
database/migrate-phase51-platform-ops-tables.sql	Step 9	Creates empty tables
database/phase16-postcheck.sql	Step 10	Read-only
database/rollback-phase48-51.sql	Rollback only	Yes — drops tables/FKs
docs/PHASE-16-RUNNER-MANIFEST.txt	Phase 16b	Reference

3.2 PHP files (Phase 16b — NOT modified yet)

File	Change required	Risk
includes/database-migrations.php	Replace getDatabaseMigrationFiles() array per manifest (52 files)	High — test on clone first
includes/database-reset.php	Verify still calls runDatabaseMigrations after reset	Low
includes/staff-repository.php	deleteStaffProfileCompletely: add staff_messages, app_notifications, push_subscriptions cleanup	Medium
admin/go-live.php	Show last migration run + postcheck status	Low
includes/go-live.php	Optionally invoke phase16 postcheck query	Low

3.3 Files NOT touched in Phase 16

All admin/*.php UI, staff-app.php, api/* (except future health), apply/admin/*, assets/*, config.php on server.

4. Exact Migration Execution Plan

Step-by-step (production)

```
STEP 1 BACKUP (see Section 6)
STEP 2 Run database/phase16-precheck.sql → save output as phase16-precheck-YYYYMMDD.txt
STEP 3 If staff table missing: run migrate-phase35 through 37 manually
STEP 4 Run any missing phase 38-47 files (idempotent CREATE/ALTER)
STEP 5 Run migrate-phase48-composite-indexes.sql
STEP 6 Run migrate-phase49-backfill-staff-id.sql
STEP 7 Re-run phase16-precheck.sql sections 2,5,6,7 — MUST be 0 orphans
STEP 8 If orphans = 0: run migrate-phase50-foreign-keys.sql
      If orphans > 0: STOP — fix data manually — do NOT run step 8
STEP 9 Run migrate-phase51-platform-ops-tables.sql
STEP 10 Run database/phase16-postcheck.sql → save as phase16-postcheck-YYYYMMDD.txt
STEP 11 Production smoke test (Section 8)
STEP 12 Wait 24h — Gate G approval before Phase 15
STEP 13 Phase 16b: deploy PHP runner update (separate deploy)
```

New tables created (phase 51)

Table	Future phase
platform_cleanup_log	P18 Cleanup Center
platform_storage_snapshots	P19 Storage Analyzer
platform_audit_runs	P20 Monthly System Audit
platform_ssl_checks	P21 Domain & SSL Monitoring
emergency_event_log	P22 Emergency Event Mode
equipment_* (3 tables)	Out of scope — dormant schema only; rental removed from roadmap v1.2

5. Exact Rollback Plan

5.1 When to rollback

- phase16-postcheck shows bad_count > 0 for orphan checks
- Nick Cave registration counts changed unexpectedly
- Admin staff list errors after migrate
- FK migration fails mid-run

5.2 Rollback procedure (ordered)

```
LEVEL 1 – Partial (phase 48-51 only, data preserved)
1. Run database/rollback-phase48-51.sql
2. Note: MySQL 8.0.19+ supports DROP INDEX IF EXISTS; older: manual DROP
3. Re-run phase16-postcheck
4. Application continues – staff_id backfill from phase 49 REMAINS (beneficial)

LEVEL 2 – Full database restore (catastrophic)
1. Maintenance mode ON (system_settings maintenance_mode=1)
2. Restore MySQL dump from Step 1 backup via cPanel/phpMyAdmin
3. Verify row counts match precheck baseline
4. Maintenance mode OFF
5. Smoke test Section 8

LEVEL 3 – PHP rollback (Phase 16b only)
1. Revert includes/database-migrations.php from git/backup
2. deploy.ps1
3. No DB change needed if SQL not re-run
```

5.3 What rollback does NOT undo

migrate-phase49-backfill-staff-id.sql population of staff_id — intentional. Level 1 rollback does not clear staff_id. Level 2 full restore does.

6. Exact Backup Procedure

6.1 Pre-migration backup (mandatory)

```
PRODUCTION (admin.olasentra.com):
1. Login → Admin → Settings → Backups
2. Click "Run full backup now" (backup-run.php)
3. Download all 3 weekly files from backups.php:
  - event-staff-database-*.sql
  - event-staff-settings-*.json
  - event-staff-site-*.zip
4. Store locally: C:\Backups\olasentra\phase16-YYYYMMDD\

LOCAL (developer machine):
powershell -ExecutionPolicy Bypass -File .\scripts\pre-deploy-backup.ps1
→ storage/backups/pre-deploy-YYYYMMDD-HHMMSS.zip

OPTIONAL cPanel:
phpMyAdmin → event_staff_system → Export → SQL → Save

RECORD:
- Backup timestamp
- File sizes
- phase16-precheck output
- Git commit hash (if any)
```

6.2 Backup verification

- SQL dump opens in text editor; contains CREATE TABLE staff
- Dump size > 100 KB (production with data)
- settings JSON contains site_name key

7. Risks & Mitigations

Risk	Severity	Mitigation
FK fail on orphan rows	Critical	precheck step 7 gate before phase 50
phase 49 creates duplicate staff	High	INSERT only WHERE email not in staff; UNIQUE on staff.email
Index already exists	Low	Runner skips duplicate; manual run may error — ignore
Long lock on staff_registrations UPDATE	Medium	Run off-peak; 36 rows = instant on current scale

equipment tables empty	None	Dormant — no development (not a rental platform)
Runner re-runs phase 35 on existing staff table	Low	CREATE TABLE IF NOT EXISTS

8. Exact Testing Procedure

8.1 Pre-migration tests (before Step 5)

#	Test	Expected
1	https://admin.olasentra.com/api/health.php	{"database":"ok"}
2	admin/staff.php?event_id=1	36 rows, all roles
3	admin/dashboard.php	Loads, pending count correct
4	register.olasentra.com index.php	Form loads
5	phase16-precheck.sql	Save output file

8.2 Post-migration tests (after Step 10)

#	Test	Expected
1	phase16-postcheck.sql	All bad_count = 0
2	SHOW INDEX idx_staff_reg_event_status	Exists
3	8 platform tables in information_schema	All present
4	staff.php event 1	Still 36 rows
5	staff-directory.php	Profiles load
6	staff-inbox.php / notifications.php	Load (FK intact)
7	job-records.php, personal-invoices.php	Load if used
8	submit.php test registration (staging email)	Creates row + staff_id set
9	scan-checkin.php	Camera page loads
10	apply-portal.php SSO	Bridge works

8.3 Performance verification

```
EXPLAIN SELECT COUNT(*) FROM staff_registrations WHERE event_id=1 AND status='pending';  
→ Should use idx_staff_reg_event_status after phase 48
```

8.4 24-hour monitoring (Gate G)

- No new errors in storage/logs/error.log
- api/health.php green
- No staff reports of registration issues
- Admin flash errors on dashboard

9. Effort & Approval

Work package	Effort	Owner
SQL package generation	Done	Architect
Production SQL execution	2 hours	DevOps + Admin
Phase 16b PHP runner	4 hours	PHP Engineer
deleteStaffProfileCompletely fix	2 hours	PHP Engineer
Total Phase 16	~1 week calendar	—

Gate F — Approve to execute on production:

1. Backup procedure accepted
2. SQL file list accepted
3. Rollback plan accepted
4. Testing checklist accepted
5. Maintenance window agreed (off-peak recommended)

Signature: _____ Date: _____

Phase 15 blocked until Gate G (P16 postcheck + 24h stable) is confirmed.